Significance tests (cont.…)

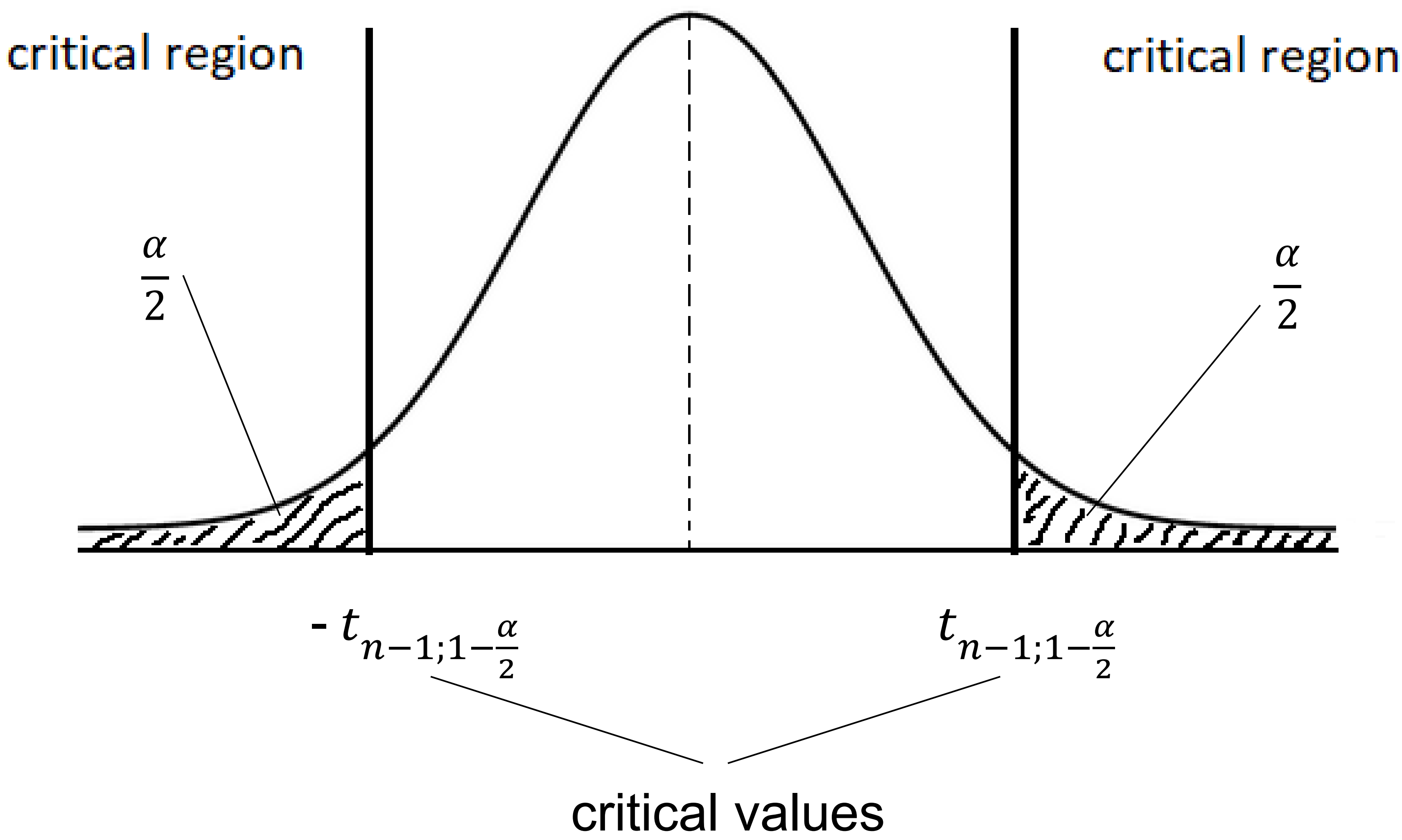
For the acceptance/critical region approach (also called the critical value approach), we need to compute the test statistic and to find the critical values (i.e., the quantiles) corresponding to a given significance level α.

For the one-sample t-tests, depending on the nature of the alternative hypothesis, we use a two-tailed or one-tailed test:

One sample *t*-test

- two-tailed test:

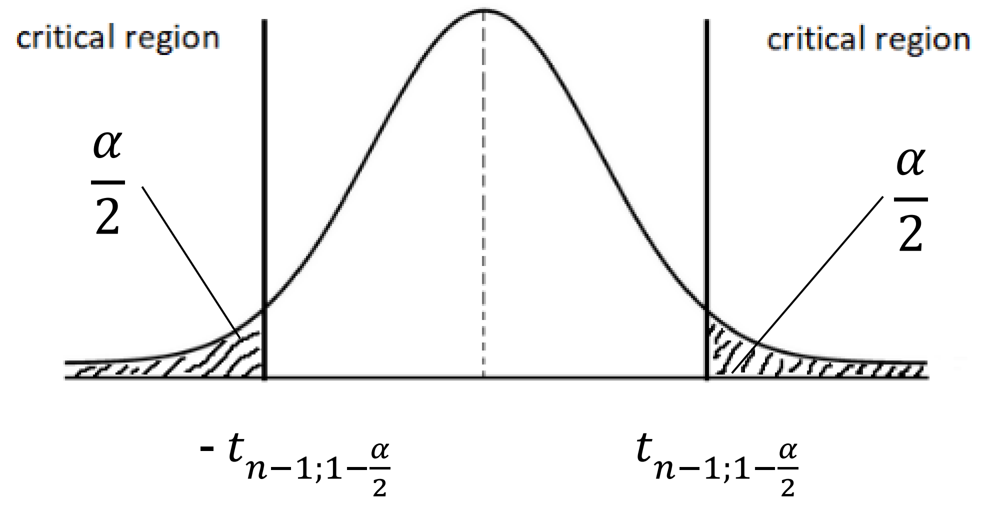
  H_0: $\{\mu = \mu_0\}$ H_a: $\{\mu \neq \mu_0\}$



One sample *t*-test

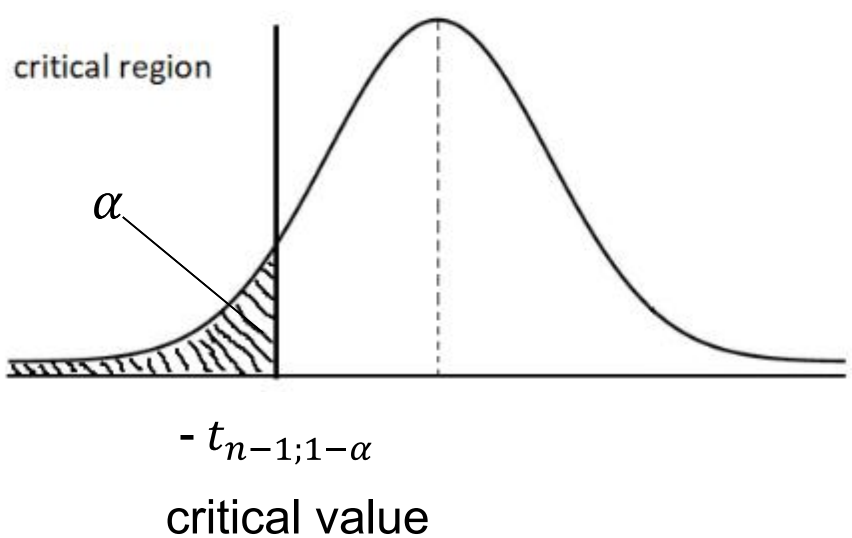
- two-tailed test:

  H$_0$: $\{\mu = \mu_0\}$ H$_a$: $\{\mu \neq \mu_0\}$



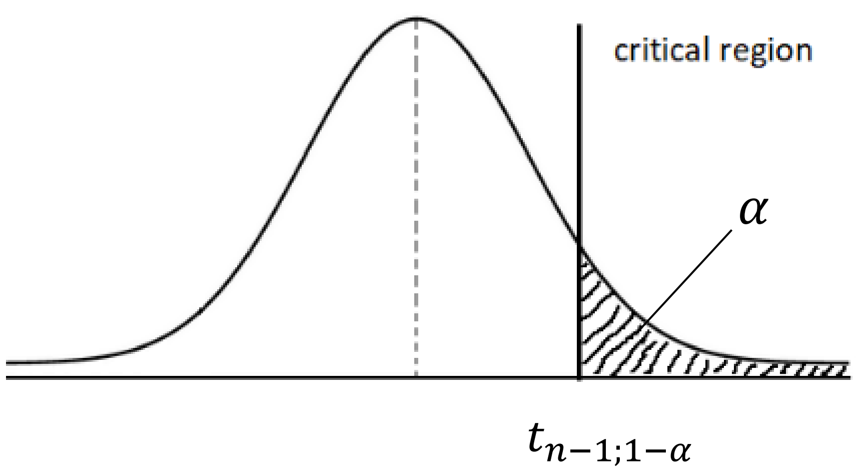
- left one-tailed test:

  H$_0$: $\{\mu = \mu_0\}$ H$_a$: $\{\mu < \mu_0\}$



- right one-tailed test:

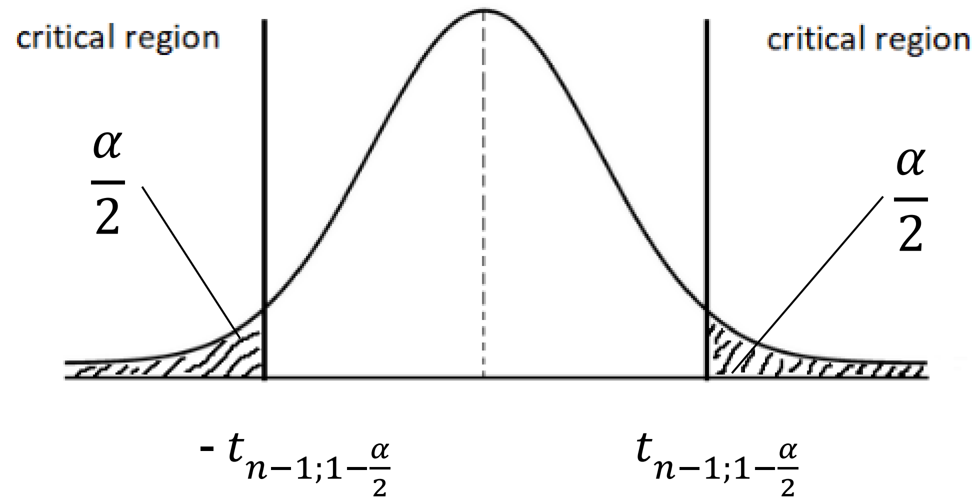
  H$_0$: $\{\mu = \mu_0\}$ H$_a$: $\{\mu > \mu_0\}$



One sample *t*-test

- two-tailed test:

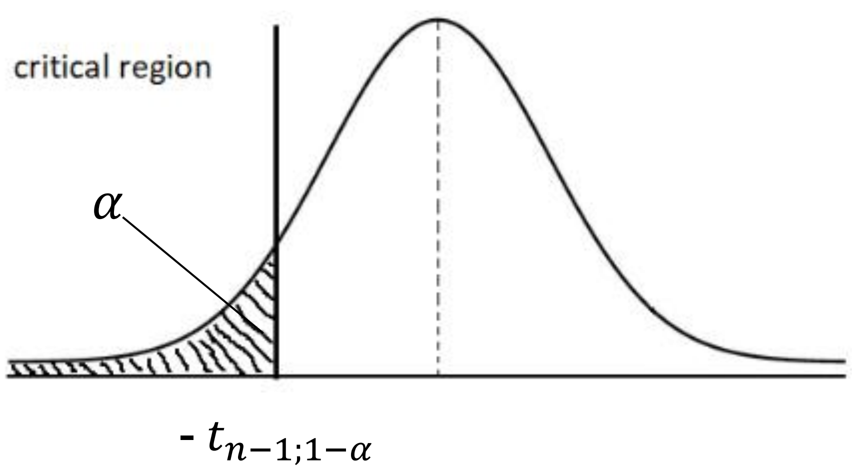
H_0: $\{\mu = \mu_0\}$ H_a: $\{\mu \neq \mu_0\}$



In all cases, the test statistic is $t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$.
If t ∈ critical region then H_0 is rejected; otherwise, we fail to reject H_0.

- left one-tailed test:

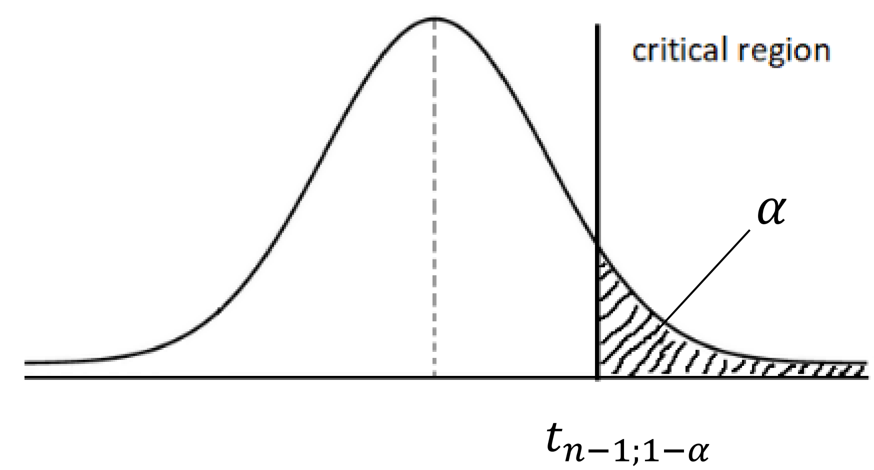
H_0: $\{\mu = \mu_0\}$ H_a: $\{\mu < \mu_0\}$



critical value

- right one-tailed test:

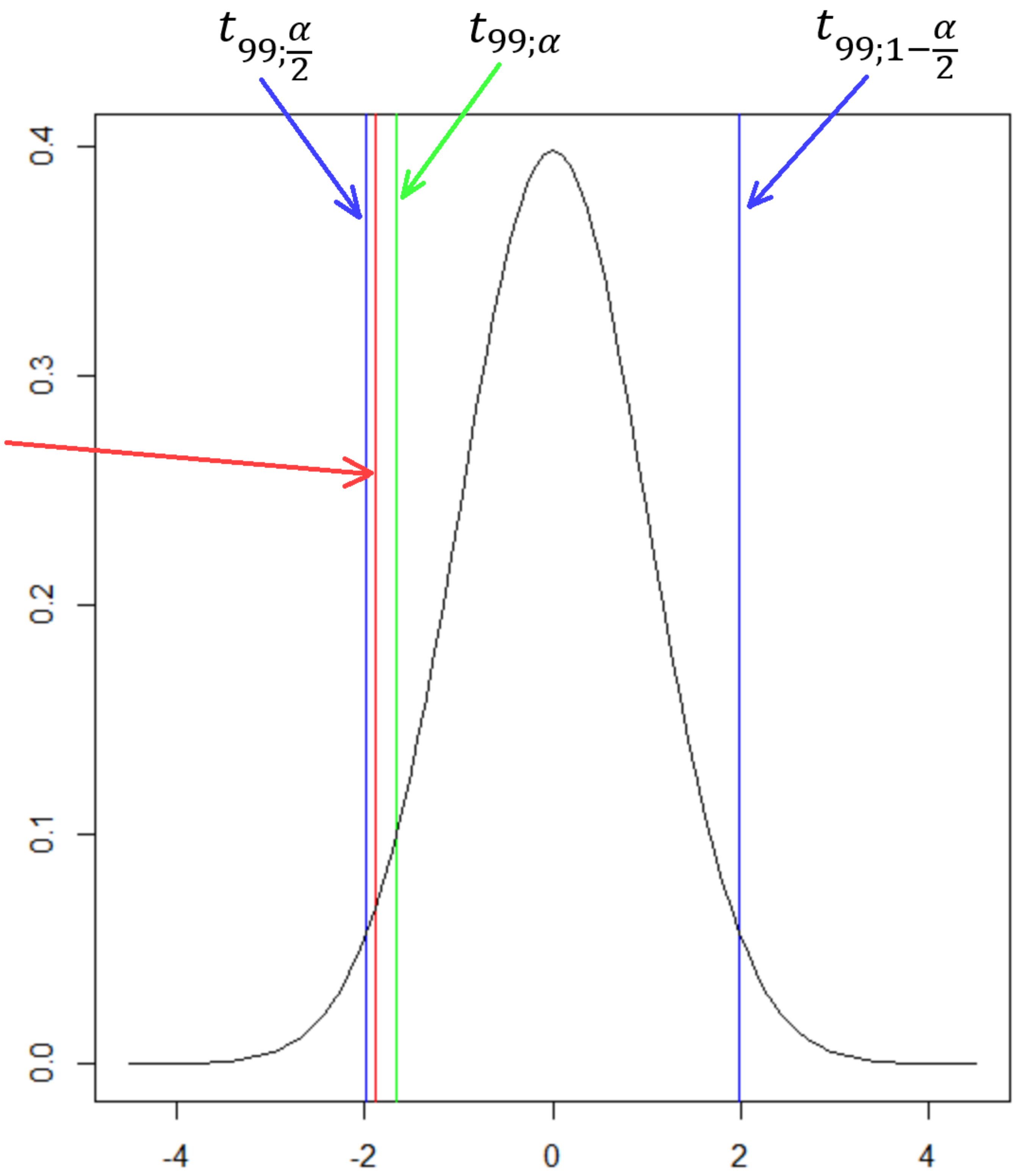
H_0: $\{\mu = \mu_0\}$ H_a: $\{\mu > \mu_0\}$



One sample *t*-test – one-tailed vs. two-tailed

α=0.05 df=99

$$t = \frac{\sqrt{n}\,(\bar{x} - \mu)}{s}$$




```
x<-rnorm(100,2,2)

a=t.test(x,alternative="less",mu=2.13)

b=t.test(x,alternative="two.sided",mu=2.13)
```

a$statistic is -1.877531

b$statistic is -1.877531

The *p*-value

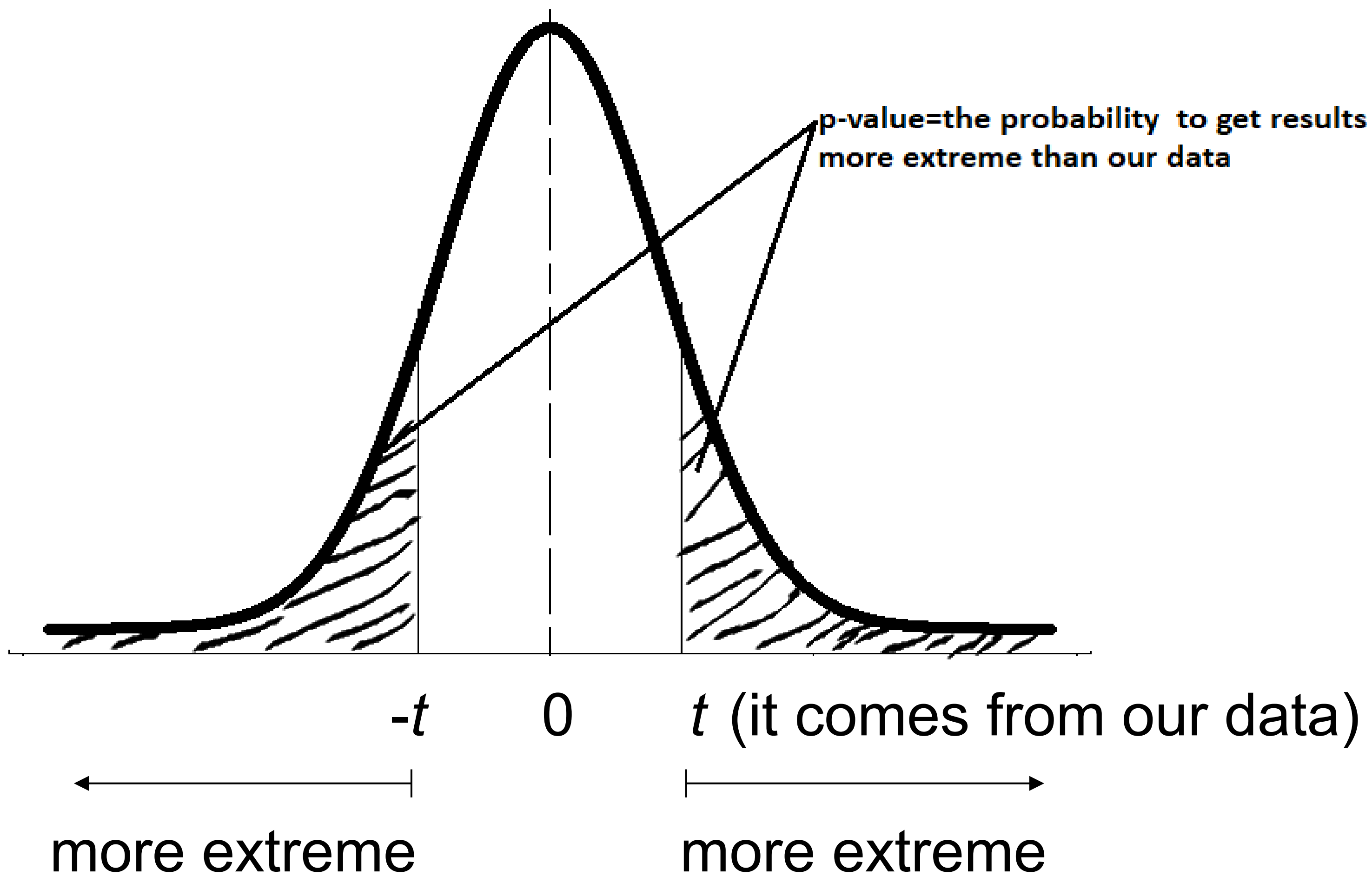
It gives the same result as the critical value approach when testing hypothesis (it is used by most statistical software).

The *p*-value is the probability that, under the assumption that the null hypothesis is true, we obtain a result " more extreme" than the one observed in our data.

p-values are triggers to decide when to reject the null hypothesis and when to fail to reject it.

The *p*-value

For two-tailed *t*-test, let’s assume that the sample data give a positive test statistic *t*.



The p-value

If p-value $< \alpha$ then the test statistic is "extreme enough" to reject H_0.

If p-value $\geq \alpha$ then we fail to reject H_0.

Obs. The condition p-value $< \alpha$ is equivalent to $t \in$ critical region of the test.

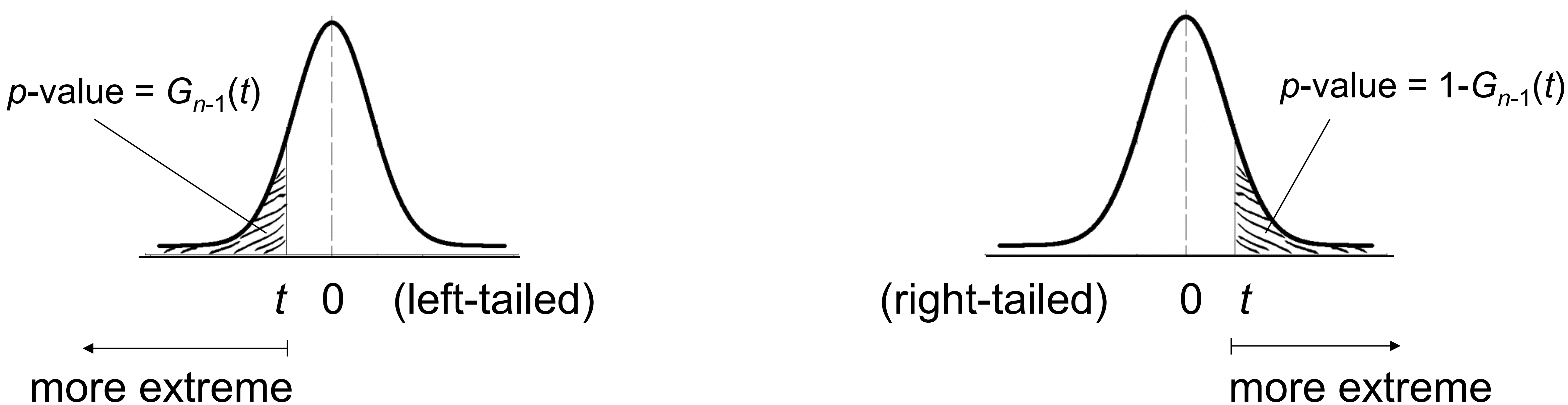
For the two-tailed t-test, the p-value is as follows:

$$p = \begin{cases} 2G_{n-1}(t), & if\ t \leq 0 \\ 2\left(1 - G_{n-1}(t)\right), & if\ t > 0 \end{cases}$$

where $t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$ and G_{n-1} is the distribution function for t_{n-1}.

The p-value

For the left one-tailed t-test, the p-value is $G_{n-1}(t)$ and for right one-tailed t-test, $1\text{-}G_{n-1}(t)$.



Obs. The p-value approach lets us choose easily the significance level value.

The *F*-test

For the variances of two normal distributions:

$X_1,\ldots,X_n$ *i.i.d.* $N(\mu_1,\sigma_1^2)$

$Y_1,\ldots,Y_m$ *i.i.d.* $N(\mu_2,\sigma_2^2)$

H_0: $\{\sigma_1^2 = \sigma_2^2\}$ against H_a: $\{\sigma_1^2 \neq \sigma_2^2\}$ (two-sided)

(or one-sided alternative
H_a: $\{\sigma_1^2 > \sigma_2^2\}$ or H_a: $\{\sigma_1^2 < \sigma_2^2\}$).

The *F*-test

Let $S_1^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}\,)^2$, $S_2^2 = \frac{1}{m-1}\sum_{i=1}^{m}(Y_i - \bar{Y})^2$

It can be proven that

$$\left.\begin{array}{l} \frac{(n-1)S_1^2}{\sigma_1^2} \sim \chi^2(n-1) \\ \frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2(m-1) \\ \text{and they are independent} \end{array}\right\} \Rightarrow$$

$$\Rightarrow \quad \frac{\frac{1}{n-1}}{\frac{1}{m-1}} \cdot \frac{\frac{(n-1)S_1^2}{\sigma_1^2}}{\frac{(m-1)S_2^2}{\sigma_2^2}} = \frac{S_1^2}{S_2^2} \cdot \frac{\sigma_2^2}{\sigma_1^2} \sim F_{n-1,m-1}$$ (*F* distribution).

The *F*-test

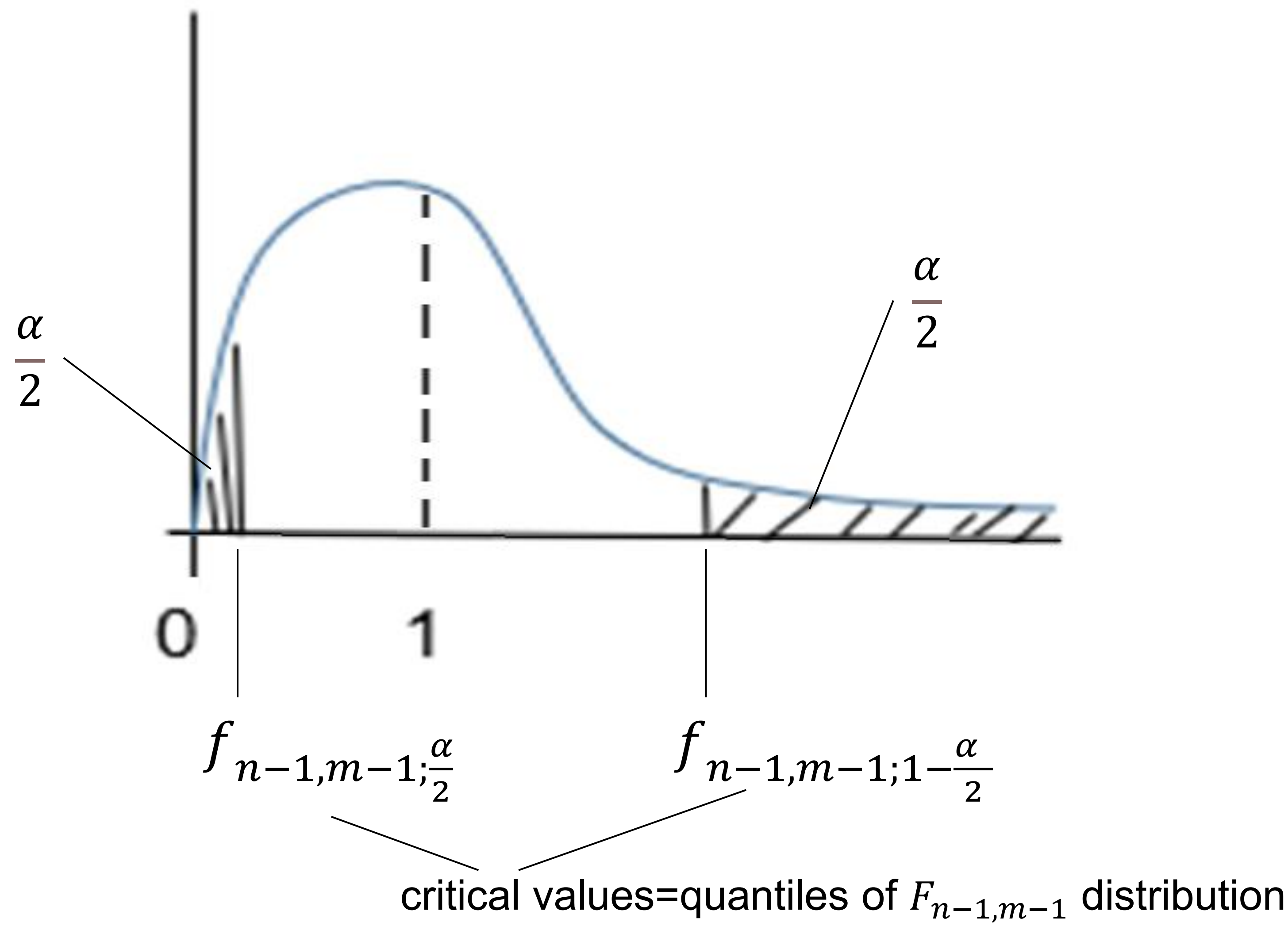
If we denote by $\gamma = \frac{\sigma_1^2}{\sigma_2^2}$, then our testing problem can be written as

H_0: $\{\gamma = 1\}$ H_a: $\{\gamma \neq 1\}$

The test statistic is $\frac{S_1^2}{S_2^2}$. Under the null hypothesis ($\gamma = 1$), $\frac{S_1^2}{S_2^2}$ has a $F_{n-1,m-1}$ distribution.

For the significance level α, the critical values of the two-sided *F*-test are:

The *F*-test



The acceptance region of H_0:

$$W_{n,m;1-\alpha}(\gamma = 1) = \{(x_1, \ldots, x_n, y_1, \ldots, y_m) | f_{n-1,m-1;\frac{\alpha}{2}} \leq \frac{s_1^2}{s_2^2} \leq f_{n-1,m-1;1-\frac{\alpha}{2}}\}.$$

The probability of type I error $P(H_a|H_0)=\alpha$.

The *F*-test

The *p*-value $= \begin{cases} 2S_{n-1,m-1}\left(\frac{s_1^2}{s_2^2}\right), & if \ \frac{s_1^2}{s_2^2} < 1 \\ 2\left(1 - S_{n-1,m-1}\left(\frac{s_1^2}{s_2^2}\right)\right), & if \ \frac{s_1^2}{s_2^2} \geq 1 \end{cases}$

$S_{n\text{-}1,m\text{-}1}$ is the $F_{n\text{-}1,m\text{-}1}$ distribution function.

<u>Obs</u>. *p*-value < *α iff* $\frac{s_1^2}{s_2^2} \in \overline{W}_{n,m;1-\alpha}(\gamma = 1)$.

The operating characteristic function:

$$\text{OC}(\gamma) = P_\gamma((X_1, \ldots, X_n, Y_1, \ldots, Y_m) \in W_{n,m;1-\alpha}(\gamma = 1))$$

$$= P_\gamma\left(f_{n-1,m-1;\frac{\alpha}{2}} \leq \frac{S_1^2}{S_2^2} \leq f_{n-1,m-1;1-\frac{\alpha}{2}}\right)$$

$$= P_\gamma\left(\frac{1}{\gamma} f_{n-1,m-1;\frac{\alpha}{2}} \leq \underbrace{\frac{S_1^2}{S_2^2}\frac{1}{\gamma}}_{\sim F_{n-1,m-1}} \leq \frac{1}{\gamma} f_{n-1,m-1;1-\frac{\alpha}{2}}\right)$$

$$= S_{n-1,m-1}\left(\frac{1}{\gamma} f_{n-1,m-1;1-\frac{\alpha}{2}}\right) - S_{n-1,m-1}\left(\frac{1}{\gamma} f_{n-1,m-1;\frac{\alpha}{2}}\right).$$

Two sample *t*-test

For the means of two normal distributions:

$$X_1,\ldots,X_n \textit{ i.i.d. } N(\mu_1,\sigma_1^2)$$

$$Y_1,\ldots,Y_m \textit{ i.i.d. } N(\mu_2,\sigma_2^2)$$

1. <u>The case of equal variances</u> $\sigma_1^2 = \sigma_2^2 = \sigma^2$

H$_0$: $\{\mu_1 = \mu_2\}$ H$_a$: $\{\mu_1 \neq \mu_2\}$ (or the one-tailed alternatives $\{\mu_1 > \mu_2\}$ or $\{\mu_1 < \mu_2\}$)

Two sample *t*-test

It follows that $\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \sigma^2\left(\frac{1}{n} + \frac{1}{m}\right)\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim N(0,1)$.

As $\frac{1}{\sigma^2}\left((n-1)S_1^2 + (m-1)S_2^2\right) \sim \chi^2(n+m-2)$, it follows that

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{1}{n+m-2}\left(\frac{1}{n} + \frac{1}{m}\right)\left((n-1)S_1^2 + (m-1)S_2^2\right)}} \sim t_{n+m-2}.$$

If we denote $\delta = \mu_1 - \mu_2$ then the hypothesis become:

H_0: $\{\delta = 0\}$ against H_a: $\{\delta \neq 0\}$ (for the two-tailed case).

Two sample t-test

The test statistic is

$$t = \frac{\bar{X}-\bar{Y}}{\sqrt{\frac{1}{n+m-2}\left(\frac{1}{n}+\frac{1}{m}\right)\left((n-1)S_1^2+(m-1)S_2^2\right)}}.$$

Under the null hypothesis assumption, $t \sim t_{n+m-2}$.

The test is built now in a similar way to the one-sample t-test (prev. course, slides 20-).

Thus, the acceptance region of H_0 is

$$W_{n,m;1-\alpha}(\delta\ = 0) = \left\{(x_1, \dots\dots, x_n, y_1, \dots\dots, y_m) \mid -t_{n+m-2;1-\frac{\alpha}{2}} \leq t \leq t_{n+m-2;1-\frac{\alpha}{2}}\right\}.$$

Two sample *t*-test

$\mathrm{P}(H_a|H_0) = \alpha$

The *p*-value = $\begin{cases} 2G_{n+m-2}(t), & if\ t \le 0 \\ 2\big(1 - G_{n+m-2}(t)\big), & if\ t > 0 \end{cases}$

The operating characteristic function is:

$$\mathrm{OC}(\delta) = P_\delta(-t_{n+m-2;1-\frac{\alpha}{2}} \le t \le t_{n+m-2;1-\frac{\alpha}{2}})$$

$$= P_\delta(-t_{n+m-2;1-\frac{\alpha}{2}} - \Delta \le \underbrace{t - \Delta}_{\sim t_{n+m-2}} \le t_{n+m-2;1-\frac{\alpha}{2}} - \Delta)$$

$$= G_{n+m-2}(t_{n+m-2;1-\frac{\alpha}{2}} - \Delta) - G_{n+m-2}(-t_{n+m-2;1-\frac{\alpha}{2}} - \Delta),$$

where $\Delta = \dfrac{\delta}{\sqrt{\frac{1}{n+m-2}\left(\frac{1}{n}+\frac{1}{m}\right)\left((n-1)S_1^2+(m-1)S_2^2\right)}}$.

Two sample *t*-test

2. <u>The case of unequal variances</u> $\sigma_1^2 \neq \sigma_2^2$

The test statistic is $t' = \dfrac{\bar{X}-\bar{Y}}{\sqrt{\frac{S_1^2}{n}+\frac{S_2^2}{m}}}$

If H_0 is true, then $t' \sim t_{df}$,

where $df = \dfrac{\left(\frac{S_1^2}{n}+\frac{S_2^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_1^2}{n}\right)^2+\frac{1}{m-1}\left(\frac{S_2^2}{m}\right)^2}$ (this is Welch approximation for the degrees of freedom).

Using now the quantile $t_{df;1-\frac{\alpha}{2}}$,the test is similar with the case of equal variances.

The paired two sample *t*-test

$(X_1, Y_1), \ldots, (X_n, Y_n) \sim i.i.d.\ N(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix}.$$

Denote $\delta = \mu_x - \mu_y$

H_0: $\{\delta = 0\}$ against H_a: $\{\delta \neq 0\}$.

The paired two sample *t*-test

Let $d_i = X_i - Y_i \, , i = \overline{1, n}$

$$\left.\begin{aligned} \sigma_{xy} &= E\big((X - \mu_x)(Y - \mu_y)\big) = E(XY) - \mu_x \, \mu_y \\ \\ \text{Var}(X - Y) &= E\big((X - Y)^2\big) - [E(X - Y)]^2 \\ &= E(X^2 + Y^2 - 2XY) - (\mu_x - \mu_y)^2 \\ &= \underbrace{E(X^2) - \mu_x^2}_{\sigma_x^2} + \underbrace{E(Y^2) - \mu_y^2}_{\sigma_y^2} - 2E(XY) + 2\,\mu_x \, \mu_y \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \text{Var}(X - Y) = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}.$$

The paired two sample *t*-test

Thus, $d_1, \dots, d_n \sim N\left(\delta, \sigma_d^2\right)$, where $\delta = \mu_x - \mu_y$,

$$\sigma_d^2 = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$$

The estimators for δ and σ_d^2 are:

$$\hat{\delta} = \bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

$$\hat{\sigma} = S_d^2 = \frac{1}{n-1}\sum_{i=1}^{n} \left(d_i - \bar{d}\right)^2$$

The test is built in a similar way to the one-sample *t*-test.

We have that $\frac{\sqrt{n}(\bar{\boldsymbol{d}} - \delta)}{s_d} \sim t_{n-1}$.

The test statistic is $t = \frac{\sqrt{n}\,\bar{d}}{s_d}$.

The paired two sample *t*-test

The acceptance region for H_0 at the significance level *α* is:

$$W_{n;1\text{-}\alpha}=\left\{(x_1,y_1),\ldots\ldots,(x_n,y_n) \middle| -t_{n-1;1-\frac{\alpha}{2}} \leq \frac{\sqrt{n}\bar{d}}{s_d} \leq t_{n-1;1-\frac{\alpha}{2}} \right\}.$$

$\mathrm{P}(H_a|H_0) = \alpha$

$$\text{The } p\text{-value} = \begin{cases} 2G_{n-1}\left(\frac{\sqrt{n}\,\bar{d}}{s_d}\right), & if\ \frac{\sqrt{n}\bar{d}}{s_d} < 0 \\ 2\left(1 - G_{n-1}\left(\frac{\sqrt{n}\,\bar{d}}{s_d}\right)\right), & if\ \frac{\sqrt{n}\,\bar{d}}{s_d} \geq 0 \end{cases}$$

The paired two sample *t*-test

The operating characteristic function is:

$$\mathrm{OC}(\delta) = P_\delta\left(-t_{n-1;1-\frac{\alpha}{2}} \le \frac{\sqrt{n}\,\bar{\boldsymbol{d}}}{s_d} \le t_{n-1;1-\frac{\alpha}{2}}\right)$$

$$= P_\delta\left(-t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}\,\bar{d}}{s_d} \le \underbrace{\frac{\sqrt{n}(\bar{\boldsymbol{d}}-\delta)}{s_d}}_{\sim t_{n-1}} \le t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}\,\bar{d}}{s_d}\right)$$

$$= G_{n-1}\left(t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}\,\bar{d}}{s_d}\right) - G_{n-1}\left(-t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}\,\bar{d}}{s_d}\right).$$